Urban Transformations

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Walking through the microclimate: potential and limits of two participatory thermal walks

Hala Zouad[1,2*] and Nacima Baron[1*]

*Correspondence:
Hala Zouad
zouadhala@gmail.com
Nacima Baron
nacimabaron@gmail.com
[1]Université Gustave Eiffel, Champs-sur-Marne, France
[2]Universitat Politècnica de València, Valencia, Spain

Abstract

Thermal walks have emerged as a participatory experimentation method that integrates both quantitative and qualitative approaches. Such experimentations are designed to capture real-time thermal perceptions in relation to diverse urban features, such as greenery, urban design, and morphology. However, in practice, these activities often prioritize quantitative methodologies, leaving the qualitative dimension underexplored. This article aims to bridge this gap by incorporating findings from a literature review and a case study focusing on the qualitative dimension of thermal comfort. Both enable authors to analyse the conditions and limitations of experimentation with urban heat. The paper concludes with a critical and reflective evaluation of the case study experiment and offers recommendations for urban planners to address the spatial, sensory, and social complexities of urban heat.

Science highlights

- Thermal walks cover a range of disciplinary approaches and are expanding their methodologies and goals.
- Heat perception data are embedded in much broader sensorial and social valuations of place.
- Thermal walks open a process of reflexive learning about heat and the ways to cope with it, individually and collectively.

Policy and practice recommendations

- Recruit participants from diverse socio-demographic backgrounds to capture pluralistic perceptions and meanings of heat.
- Foster collaborative multi-stakeholder design involving academia, local governments, and community organizations to ensure thermal walk itineraries and research reflect community priorities.
- Conduct multiple thermal walks to deepen local community engagement with heat adaptation planning and preparedness.
- Refine climate adaptation solutions (e.g., greening interventions) through participatory approaches such as community-driven planting projects in the areas most in need.

Keywords Thermal walks, Participatory experimentation, Thermal perceptions, Thermal comfort, Urban heat

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Introduction

Urban heat is a silent killer (UNDRR 2025). Even when it is not fatal, it still affects our daily lives. Yet urban heat is not simply an inevitable consequence of global climate change. It is also shaped and intensified by local planning choices concerning built form, green space, and car traffic. These problems are particularly marked in disadvantaged and segregated districts, where minority communities are often concentrated. It is expected that, in the future, communities living in the most exposed districts will face increasing health risks (Ebi et al. 2021; Ballester et al. 2023). Urban heat is therefore both an environmental threat and a marker of urban inequalities and institutional neglect (Klinenberg 2015).

The social sciences have increasingly focused on the challenges of a just and sustainable collective life in tomorrow's hotter cities, questioning who has the voice and power over the design and use of hotter public spaces (Steele et al. 2023). Scholars in political economy point out that an apparatus of economic and political powers is introducing a range of "techno-solutionist" cooling interventions such as more greenery, textile-shaded areas, water streams and fountains, or white-painted soils and roofs (Kaika et al. 2023). They note that such responses are often associated with green gentrification, which refers to the integration of speculative urban projects and climate-oriented, redesigned areas. They highlight two associated risks. One can fear that heat-proofed urban environments may be reserved for the more affluent part of cities. One can also regret that the democratic energies that shape cities and communities through grassroots movements may be set aside in favor of a top-down coalition of business, science, and institutions in urban climate planning. If we want to reverse this scenario, we must start the work in the very places where the heat is most accentuated and where solutions may come late. Therefore, we must think and act with and for local communities. Participatory thermal walks may help address this risk by revealing inequalities in access to cooling resources and involving affected residents in defining local priorities. However, participation alone cannot prevent green gentrification. It must be accompanied by broader housing and social protection policies to ensure that climate improvements do not contribute to the displacement of vulnerable residents.

To date, there is limited knowledge and understanding of the physiological and cognitive perceptions of heat among local, disadvantaged communities. Heat planning shifts toward climatology and the domestic air-conditioning industry, both of which utilize the concept of thermal comfort —a concept that integrates environmental and physiological parameters (ASHRAE 55). Planners can obtain thermal comfort maps from urban climate models that account for urban morphology, wind conditions, and solar exposure. However, by construction, these maps fail to capture the reality and meaning of heat in its situational and relational context. Urban planners and residents can now utilize a tool to navigate the complexities of perceived heat. Thermal walks are gaining traction in both academic research and professional planning practice. Drawing on urban microclimatology, researchers and practitioners use them to investigate how specific groups perceive heat in particular urban settings. This disciplinary transfer raises a central question: how can planners adapt methods developed in climatology to foster participation and collective learning?

The first part presents an overview of heat perception and heat adaptation, introducing thermal walks through a comprehensive literature review. It highlights how urban

planners refine, recontextualize, and adapt microclimatological protocols through in situ experimentation. The second part focuses on implementing two thermal walks in Ivry-Port, a deprived district on the outskirts of Paris. It serves as a lens to detail the potentials and limitations of the concept of thermal walks for urban action, rather than for climate diagnosis. The third part offers recommendations for urban practitioners and outlines the remaining epistemic challenges if we, as urban scholars, aim to further "urbanize" and "commonize" the concept of thermal comfort for the benefit of the majority of metropolitan residents.

Part 1: Exploring and experimenting (with) heat?

Heat perception and heat adaptation

Urban heat can be defined as a spatial and temporal concept. Spatially, it relates to the urban heat island effect (UHI), a phenomenon in which urban areas are significantly hotter than their surrounding rural areas (Oke 1982). Temporally, urban heat includes short-term events, such as hot days and nights, as well as longer periods, including heatwaves and seasonally warm conditions in spring and autumn (Cheval et al. 2024). Understanding what heat means for urban dwellers is complex because it is embodied in each of our bodies and minds. The thermal experience is multisensory and shaped by various stimuli, which are processed through our sensory perception (Liu et al. 2025).

Heat perception encompasses synesthetic, psychological-emotional, physiological, and behavioral dimensions (Fig. 1). There is a permanent interplay between heat perceptions (Fig. 1, left) and our responses to heat (Fig. 1, right). Heat perception involves multiple dimensions that together shape the human experience of thermal conditions. The synesthetic dimension integrates sensory inputs from touch, sight, smell, and hearing, providing texture and context to thermal experiences. The psychological-emotional dimension encompasses the range of affective responses to heat, from discomfort to pleasure, which are influenced by situational factors.

Thermal perception is shaped by multiple factors, such as awareness, psychoterrain, exposure history, and alliesthesia. Each of these factors will be examined in turn. Heat awareness means that we can fear the risks associated with heat, such as its health impacts. Psychoterrain designates the mental framework shaped by continuous exposure to urban heat. This psychoterrain not only informs how we interpret heat, but it also conditions our responses. Exposure history refers to our memory of former heat experiences. Past exposure shapes our expectations of future heat events. Responses to heat involve both physiological changes and emotional reactions. Bodily responses include sweating or changes in heart rate. They can occur in real-time, but they may also exhibit delays or lags, adding complexity to the experience of heat (Kelly 2011). Emotional

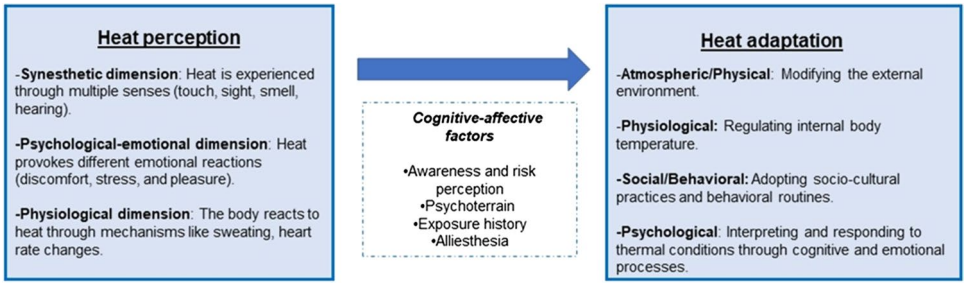


Fig. 1 How heat perceptions, cognitive, and bodily heat adaptation interact

responses include positive or negative alliesthesia, which refers to the (dis)pleasure following a thermal stimulus (Dzyuban et al. 2022). Emotions significantly influence how people engage with and adapt to heat in urban environments (Rony and Alamgir 2023).

Urban heat brings discomfort in everyday city life, and we adapt at the individual, community, and city levels. Heat adaptation has a specific meaning in climatology and covers a range of facts. Sweating is an example of unconscious, indirect, and involuntary body adaptation. Yet each individual can seek adaptations in mobility (such as switching to the shaded sidewalk), in work habits (schedules and location), in clothing, and so on. We consider four main elements of heat adaptation, as outlined by Nikolopoulou and Steemers (2003) and Rahm (2023). However, we emphasize that various factors, including age, gender, social, psychological, and religious aspects, can either facilitate or impede heat adaptation. Two examples illustrate this point. Adapting clothing to temperature changes in public spaces is part of both physiological and social behavior, as cultural conditions can influence these reactions, which vary by gender and context. Adapting depends on the individual's actions and circumstances, and not only on their capacity. For example, people can choose a shaded place if they are able to walk, but waiting at an unsheltered bus station limits their options for heat adaptation.

Thermal walks, a case of disciplinary transfer

The interaction between an embodied experience and the thermo-spatial environment, mediated through walking, forms the foundation of thermal walks. Thermal walks have been mobilized successively in four scientific contexts since the end of the 1990s. An evolutionary perspective of thermal walks reveals not only a variety of definitions, methods, and goals, but a transfer and relative merging of this protocol from the science of climate toward an urban participatory dimension.

Thermal walks have emerged in climatology to understand pedestrian thermal comfort in outdoor environments. Thermal walks have been explored in the academic literature for more than three decades. Nagara et al. (1996) combined subjective thermal evaluations of pedestrians with real-time physico-thermal measurements along a major artery in Japan. The authors reported that perceptions were shaped not only by immediate conditions but also by exposure history and visual cues (e.g., the presence of a river). This insight opened a new path, and subsequent researchers (Nikolopoulou et al. 2001; Vasilikou and Nikolopoulou 2020; Dzyuban et al. 2022) have further explored the relationship between subjective thermal perceptions and objective measurements, considering factors such as urban design, morphology, green spaces, microclimatic variations, and human adaptation.

In a broad sense, thermal walks now consist of organizing a collective "stop-and-go" protocol to gather physico-thermal information from individuals, as a complement to measurements conducted at each stop. Participants respond to questionnaires and undergo behavioral observations or body instrumentation while microclimate measurements are conducted. Throughout the 2000s and 2010s, the protocol was refined and enriched. Additional environmental and urban design factors were included as criteria for understanding micro-local climates, such as solar radiation and the effects of building height and volume. Climate indices that assess and forecast thermal comfort under varying environmental conditions have been developed. The modified mPET, a more complex model, was preferred over the original Physiological Equivalent Temperature

(PET). Yet urban microclimatologists' focus remains unchanged: to define optimal conditions for human comfort based on biological responses to the physical environment.

Later, this approach gained momentum in participatory urban planning. Since 2020, urban planners have been interested in the transfer of thermal walks into a broad series of mobile participatory urban walks. Sometimes called sense walking, climate walking, or weather walks, they all belong to the emerging field of walking public pedagogies. In this transfer from climatology to urban disciplines and from academia-led fieldwork to urban experimentation, the purpose of the walk has been partially reframed. Urban thermal walks adopt both a climate and a social perspective. The aim is not only to record the impact of physical aspects of urban design on heated bodies and minds. For urban planners, thermal comfort extends beyond temperature, humidity, and urban morphology and must also account for its impact on individual perception. Thermal walks now have their place in participatory mobile methods with a vision of democratizing climate policies. Their broad goal is twofold. First, they examine how physiological, physical, environmental, and social factors interact. In this sense, they are part of a toolkit that helps planners navigate the complexities and perhaps the ambiguities of urban heat as it is perceived and experienced locally. Second, they can serve local governments that want to put equity into climate adaptation policies. Communities may have a say in the types of planning and urban management responses that can be provided (e.g., more public-space shelters, more trees, but specifying where, how, and so on).

Thermal walks

Even if climatologists and planners have different approaches to defining and framing the aims of a thermal walk, the experimental protocols developed across various experiments span both climatology and planning. A literature review was conducted on November 21, 2024, and two databases were searched (Google Scholar and ScienceDirect). The keyword for the search was "thermal walk." The exclusion criteria were based on language (only English) and did not limit the year range. Parallel searches were also conducted in Scopus and Web of Science, but they did not yield a sufficient number of relevant studies matching our focus on thermal-walk experimental protocols. Consequently, ScienceDirect was retained as the main database, complemented by backward snowballing from the reference lists of the selected papers. Google Scholar was used as a complementary search tool. We then conducted an in-depth analysis of 9 papers, selected based on the distinctiveness of their experimental protocols (Table 1).

Thermal walks can thus be understood as a boundary object (Caccamo 2023), increasingly mobilized across disciplinary perspectives. The main takeaways from the literature review are as follows:

- Thermal walks are conducted across diverse urban spaces, morphologies, and contexts. They have been implemented in different types of urban fabric, including city centers with historical monuments, commercial areas, waterfronts, and riversides, in both large and small cities.
- Thermal walks operate at different spatial scales. They can focus on individual streets or campuses, extend across an entire district, or encompass multiple districts.
- Thermal walks are conducted across different seasons and thermal conditions. They have been used in winter to investigate sensitivity to cold and during hot periods to address urban heat. However, thermal walks are not necessarily associated with

Table 1 Comparison of 9 thermal walks in scientific literature

Authors	Spatial context and urban features	Research method	Added value of the article
Ali Smail et al. (2024)	Historical site in Algier (Algeria)	Questionnaires + measurement	Thermal pleasure and fatigue sensation of pedestrians
Yanha Li et al. (2024)	Green axis of parks and gardens of 5.5 km in Beijing (China)	Positive Affect and Negative Affect Schedule Questionnaire + measurement	Walking status (in motion or standing), ambiance (thermal, visual, and acoustic), and emotions
Dzyuban et al. (2022)	Disadvantaged neighborhood with high public housing/degraded infrastructure in Phoenix (US)	Questionnaire + meteorological measurement	mPET and PET comparison; Alliesthesia
Barbosa and Klok (2020)	Former reconverted naval base	Questionnaire + meteorological measurement	Thermal walk as an experiential learning and awareness-raising tool.
Vasilikoulou and Nikolopoulou (2013)	A centric historic sequence (500 m), London (UK) and Rome (Italy)	Environmental and human monitoring	Sequential thermal diversity (the transition between urban forms (open, closed), urban geometry (aspect ratio, sky view factor)
Vasilikoulou and Nikolopoulou (2015)	Heavy traffic street in Paris, France	Microclimatic monitoring and simultaneous people surveys	Thermal notation as a graphic representation to juxtapose objective and subjective data in the same diagram
Peng et al. (2022)	Waterfronts in Rome, Italy	Smart wearables to detect skin conductance levels + TSV questionnaires and measurements	Introducing the physiological data and combining the psychogeography and thermal walk protocols for obtaining physical-physio-psycho datasets
Xie et al. (2022)	Campus of Tsinghua University in Beijing, China	Back-type microclimate equipment + questionnaire	The influence of short-term thermal history and the sequence of thermal exposure
Nagara et al. (1996)	Business district of Osaka, Japan	Questionnaire and scientific equipment (thermometer, hygrometer)	Thermal comfort studied in indoor and outdoor settings, considering the physiological aspect and the visual impact of places

extreme weather conditions. Spring and autumn may also provide particularly relevant conditions for investigating thermal experience, as moderate and warm temperatures are associated with a wider range of urban practices and may therefore reveal a broader diversity of thermal perceptions. Moreover, conducting walks in July and August may affect participant recruitment, as these months coincide with the holiday period and some residents may temporarily leave the district.

- Thermal walks increasingly function as participatory projects. When embedded in action-research programs involving public and community stakeholders, such as municipalities and NGOs, their protocols tend to adopt a stronger participatory orientation. The literature indicates that these protocols may incorporate various questionnaires that assess emotions, sensory experiences, and perceived thermal comfort. Elected officials may also participate in the walks, while preparatory and follow-up activities can include information sessions, feedback meetings, festivals, cultural events, or health-prevention campaigns.

The two Ivry-Port thermal walks nevertheless provide a temporally bounded account of thermal experience and do not capture seasonal variations, particularly cold and inclement winter conditions. Weather-related barriers to walking are not limited to hot conditions. In winter cities, indoor pedestrian environments can support safe and comfortable walking during inclement weather (Balsas 2021). Although this research concerns a

different climatic and spatial context, it shares with thermal walks a concern for adapting the built environment to support walking and health. Further research could therefore compare thermal walking experiences and methods across seasons and between indoor and outdoor environments.

Part 2. Conducting two thermal walks at Ivry-Port

Ivry-Port site and urban context

Ivry-sur-Seine (80,000 inhabitants) is a long-lasting communist-led municipality directly adjacent to the southern limit of Paris. The Ivry-Port district (6500 inhabitants) corresponds to a post-industrial waterfront of the Seine River. It is the poorer part of this city. The poverty rate, defined as the percentage of the population living below the 60% income threshold, is 42% in 2024, according to Sig.ville.gouv.fr, the French geographic information system run by the Ministry of Cities. The neighborhood features a blend of ruined industrial warehouses and 20th-century streets, where affordable rental housing was provided for workers in 3- to 5-story brick-and-concrete buildings. In terms of urban greening, the municipality, which was controlled by the communist party for almost a century, followed a traditional planning vision. The main avenues were lined with trees, but as car traffic increased and parking demand grew, some of these trees were partially removed. Moreover, greening was traditionally considered an amenity primarily for mothers and children. Several small parks with playgrounds are located near some primary municipal schools. Football pitches were also provided for young people. Although some green spaces remain in school grounds, many playing surfaces have been replaced with artificial turf to accommodate more intensive use and reduce maintenance costs. Taken together, these features make Ivry-Port emblematic of the legacy of the Parisian "red belt", where an inherited but shrinking network of green spaces and public spaces is being reconfigured amid managed gentrification, with the aim of limiting the displacement of working-class residents while accommodating new populations.

Another feature makes Ivry-Port particularly relevant for exploring perceptions of heat. Due to its proximity to Paris, Ivry-Port is a highly sought-after location for real estate developers. It is undergoing a massive renovation project overseen by the still-communist municipality. Around 8000 housing units are planned on brownfield sites over the next decade. Currently, numerous blocks of new towers (ranging from 8 to 10 stories, sometimes 20) are inserted within the existing fabric, and some new urban greening typologies are also being introduced. The new blocks feature generous inner gardens, reserved exclusively for new middle-class residents. Yet, some pocket parks and one complete street are being planned for all. All in all, Ivry-Port provides a transitional social and urban landscape. Different population groups benefit unequally from urban greening. The majority still reside within the inherited spatial grid, with limited access to cooling facilities. The newcomers enjoy private green areas, and they also consider the recent provision of green public space to be somewhat connected to their new homes. Social tensions emerge when it turns to who can access the new, rare patches of open-air and green public spaces, and when and for what purposes.

Experimentation design

The authors have collaborated with the Department of Citizen Science of the Ivry-sur-Seine municipality, the Civic Center of the Ivry Port district (which is in touch with the

poorer and more disadvantaged parts of the district), and with a community collective that represents the interests of middle-class newcomers (Urbanicc). Two thermal walks have been conducted, engaging a total of 26 volunteers. The walks took place on October 2nd and 19th, 2024, following 8 months of preparation, which included a dozen meetings with stakeholders. We explain the conditions of their prefiguration in the following paragraphs and tables. The thermal walks were based on creating two itineraries within the district, with different participants in each.

The Ivry-Port walks were conceived as part of an iterative experimental process that operates through iteration and variation, two concepts central to what Ansell and Bartenberger (2016) call "Darwinian experimentation." Such an approach involves repeating and progressively refining a research procedure, while allowing certain aspects of the experiment to vary. The underlying hypothesis is that modifying the composition and characteristics of successive experiments enables learning through trial and error. Each thermal walk can thus become an opportunity for constructive exploration and learning from failure (Fuenfschilling et al. 2019).

Our experience with thermal walks predates the Ivry experimentation. It began one year earlier in the city of Valencia, Spain, under different conditions in terms of participants and urban context. The Ivry experiment was therefore reconceptualized based on both insights from the state of the art and lessons learned from this first field experience. This iterative process is consistent with urban climate experimentation, in which failure can contribute to methodological refinement and adaptive governance innovation (Bulkeley and Castán Broto 2013).

The protocol resulting from this iterative process is structured into three phases (Fig. 2). The first phase assesses the opportunity for a thermal walk, selects the study site, and establishes local partnerships. The second anchors the experiment in the local context through the appropriation of the site, the co-design of a pilot walk, and the implementation of the thermal walks. The third phase evaluates the two walks and draws methodological lessons from them. These successive steps provide a common structure

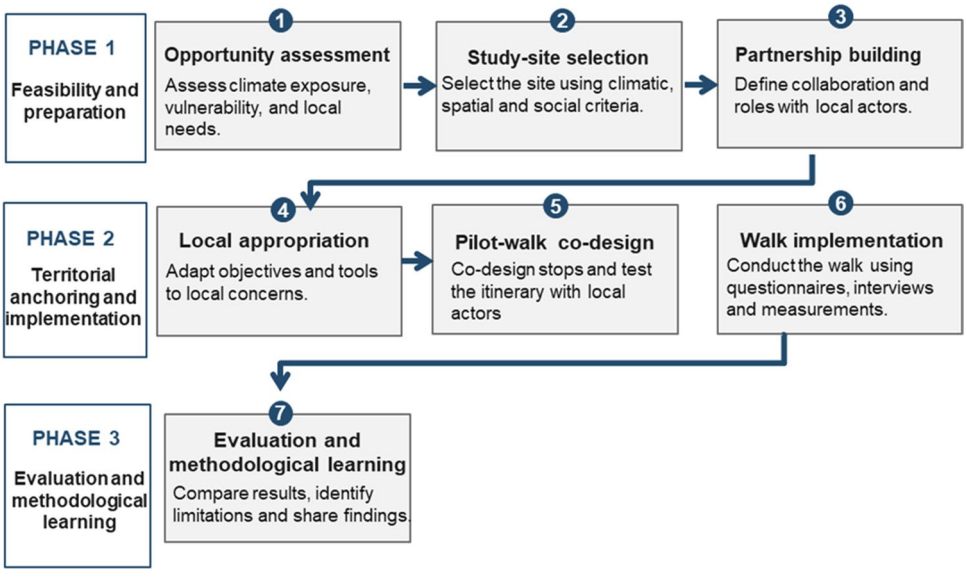


Fig. 2 Methodological protocol for the Ivry-Port thermal walks: phases and implementation steps. Source: Authors

for the thermal walks while allowing adjustments to the specific characteristics of each itinerary and its participants.

Recruitment of participants

We recruited a diverse group of participants across gender, age, origin, educational background, and social status (Table 2). The purpose was to attract people from the poorest areas of the district, as well as new residents enjoying better living conditions. The hypothesis was that the former, rooted in the inherited urban grid, suffered greater heat exposure but may have developed adaptive behavioral responses. This group was therefore expected to demonstrate greater heat tolerance over time, as repeated exposure fosters familiarity with heat and shapes thermal perception. The latter enjoyed better living conditions but may have been less accustomed to the thermal conditions of this deprived urban environment. The participation of this second group could help capture different relationships with urban green spaces and different patterns of public space use. Various communication channels were mobilized, including posting in local social media groups and placing informational posters in public areas. We distributed flyers in strategic locations, including in front of the school, near key amenities and services (such as the library and doctors' offices), and within parks and sports facilities. When the contacts were created, we held an information event at the civic community center, organized a week before each walk. During this event, we introduced climate risks and circulated heat maps and thermal comfort models. We familiarized participants with various scientific instruments, including thermal cameras, thermometers, a portable weather station, and sound level meters, enabling them to interact directly with the equipment and explore the sites' thermal characteristics. However, as discussed later in the article, the purpose of providing these instruments was not to develop a crowd science approach, but rather to foster climate citizenship. Participants were therefore encouraged to use the instruments freely, without being required to follow strict microclimatic measurement protocols.

Itineraries

Climatologists select thermal walk itineraries with great attention to environmental factors, such as the impact of natural features (rivers, parks) and the variety of built forms, materials, and colors (including albedo). Urban planners complement these considerations by attending to socio-spatial inequalities along the itineraries. They value the opportunity to understand how the environment's visual and sensory components

Table 2 The socio-demographic diversity of participants. Source: Authors

Categories	First walk	Second walk
Total number of participants	14 participants	12 participants
Gender	6 men (43%) 8 women (57%)	5 men (42%) 7 women (58%)
Age groups	24–34 years: 47%35–45 years: 33%46 years and above: 20%	24–34 years: 20%35–45 years: 50%46 years and above: 30%
Origin	**Residents of the inherited urban fabric**:– Maghrebian origin: 43%– French origin: 57% **Newcomers** (**renovated areas**): – Maghrebian origin: 25%– French origin: 75%	**Residents of the inherited urban fabric**:– Maghrebian origin: 23%– French origin: 77% **Newcomers** (**renovated areas**): – Maghrebian origin: 15%– French origin: 85%
Educational level	Primary education: 11% Secondary education: 35% Higher education: 54%	Primary education: 10% Secondary education: 25% Higher education: 65%

affect heat perception. That is why we have adopted both a microclimatological and an urban perspective. From the first perspective, we considered the width and orientation of streets in relation to the dominant winds, as well as their proximity to the river and parks. We chose to integrate an urban environment dominated by concrete, asphalt, earth, and sand. From a second perspective, we were aware of the need to integrate both inherited workers' buildings and newly built blocks, as well as 10- to 20-story towers, into the itinerary.

Both itineraries were selected in collaboration with local residents, the paper's authors, and urban stakeholders. The precise itineraries and stop choices were created by the participants during information meetings held at the civic center prior to each walk. The objective was to identify significant stops based on microclimate variations, considering both existing and newly developed urban blocks, and their relevance to everyday activities (Fig. 3). Length (in meters) and duration had to be taken into account. A consensus was reached that each walk could be completed within 1 h and 30 min to be inclusive and accommodate the walking capabilities of mothers with babies and elderly people (Fig. 4).

Unlike more rigid, climate-oriented walks, which follow fixed protocols and limit participant interaction, our approach allowed and even encouraged participants to share their perceptions. Insights were gathered through walking interviews, both with the authors and among participants themselves.

During the walks, the instruments were used in two different ways. Air temperature, relative humidity, and wind speed were continuously recorded by one researcher using a portable Kestrel meteorological station during both thermal walks. These measurements were used to characterize overall meteorological conditions and to identify microlocal variations among selected stops. Other instruments, including thermal cameras, infrared thermometers, and sound-level meters, were made available to participants throughout the walks and used at selected stops as observational and pedagogical tools. Participants recorded the readings obtained in their individual questionnaires. As these

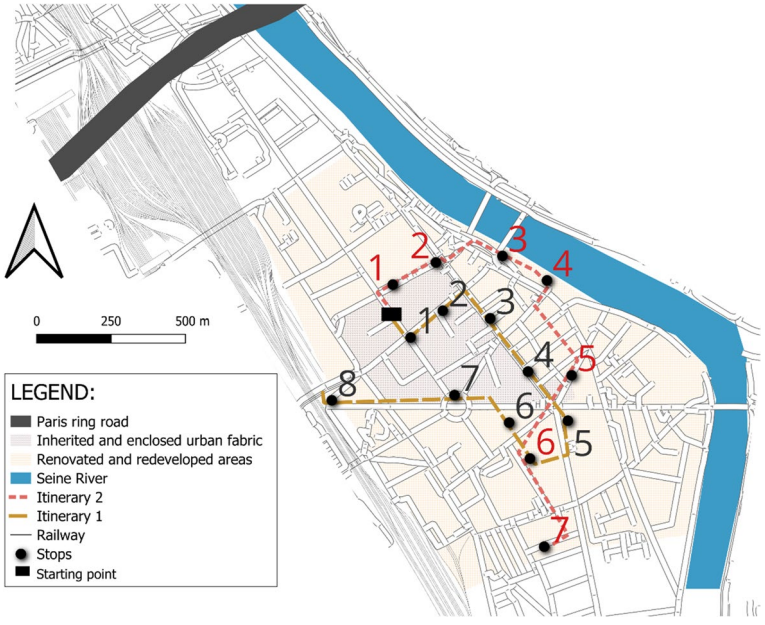


Fig. 3 Both itineraries and stops adopted during the thermal walks at Ivry-Port. Source: Authors



Fig. 4 Participants walking through Ivry-Port. Credit: Authors

readings varied according to the precise measurement point selected by each participant, they were used to support individual observation and collective discussion rather than systematic quantitative analysis. Consequently, they were not included in the quantitative dataset.

Questionnaires

In both thermal walks, two different surveys were used. The study employed both structured, stationary paper-based interviews and go-along interviews to engage participants during these spontaneous conversations. The questionnaires included individual information (such as age and gender) and behavioral aspects (such as clothing and comfort measures). They also included questions about how people psychologically responded to their thermal perceptions (i.e., how they perceived the space and what they would like to do to ease the discomfort, such as sitting down, seeking shade, or drinking water).

During the first thermal walk, at each stop, the questionnaire sought to understand how the body perceives weather elements such as wind and temperature. These subjective perceptions were complemented by meteorological measurements taken by the authors. Participants were then asked to describe their multisensory experiences, detailing what they saw, heard, and smelled. Individual thermal sensations were evaluated on a four-point scale: unpleasant, annoying, acceptable, or pleasant. Participants also reflected on their spontaneous adaptive reflexes by considering what actions they would take to cool down in that specific situation.

Building on the first questionnaire, the second thermal walk encouraged participants to envision climate-proof micro-planning solutions that could enhance comfort in hotter future conditions. At each stop, they were invited to position themselves across temporal perspectives: past, present, and future, by recalling how they felt during the last summer heat, describing their current sensations regarding the prevailing conditions in the space, and imagining possible improvements to make the place more thermally comfortable for the future.

Quantitative results

The quantitative data from the two thermal walks show fairly consistent results, justifying their combined presentation. Two main findings should be highlighted. Result 1: From a thermal perspective, the key point is that, on one hand, the study area is entirely within the large metropolitan heat island of the Paris region, which creates a certain overall homogeneity. On the other hand, there are localized and even highly localized variations in temperature and humidity. During these two autumn days, conditions were mild and above the meteorological average. While the mean temperature in October in Ivry is typically around 13 °C, we have recorded a local temperature range of 17 to 20 °C, with an average of 18.6 °C and a relative humidity of 74%. Since the capital and its surrounding areas are situated in a river valley, wind speeds are limited, averaging around 0.15 m/s. Air circulation significantly influences thermal perception. The cooling effect of the Seine's hydrographic corridor, which borders the district, was nearly imperceptible. This is due to the morphology, density, orientation, and height of the buildings. Larger and taller buildings from recent redevelopment phases restrict air flow, whereas areas still made up of former industrial wastelands or undeveloped plots allow breezes to flow more freely.

However, measurements during the thermal walks reveal distinct micro-local thermal contexts. Differences are evident between points that are very close geographically. During the first thermal walk, a temperature contrast of 2 °C was recorded between the Fauconnières Bloc (stop 2, itinerary 1, see Fig. 3) and the Vaillant-Couturier bus stop (stop 3), despite their proximity of only two minutes on foot. In the second walk, the difference increased to 3.7 °C. The warmer site was located along the Seine, at the base of recently constructed high-rise towers. Although the river environment typically exerts a cooling influence, this site recorded the highest local temperature (22.2 °C). This anomaly may be related to the strong presence of heat-retaining and polluting infrastructures, including highways, industrial zones, and a nearby cement plant with heavy truck traffic and dust emissions (stop 3, itinerary 2). In contrast, the coolest measurement (18.7 °C) was taken in the small Prudhon pocket park (Final stop 7, itinerary 2) within the renovated district, a compact, vegetated area with dense tree and shrub cover interspersed with some open ground.

Result 2: The neighbourhood is undergoing residential redevelopment. This process has produced contrasting urban morphologies that may generate different thermal experiences. The renovated area was expected to provide better thermal perceptions. However, the responses collected along itinerary 2 did not show a consistent improvement. Comfort scores ranged from 2.1 to 3.2 along the first itinerary (Fig. 5a). They ranged from 1.8 to 3.8 along the second itinerary (Fig. 5b). The second itinerary therefore showed greater variation between stops, but not systematically higher comfort. Adaptive responses were broadly comparable across the two itineraries (Fig. 5c–d). Percentages in panels c–d represent one primary adaptive response per participant. Drinking water was the most frequently reported response (57.1% and 50%). It was followed by the use of personal cooling devices (28.6% and 33.3%) and seeking shade (14.3% and 16.7%). These results suggest that the renovated urban fabric did not consistently provide greater perceived thermal comfort than the older urban fabric.

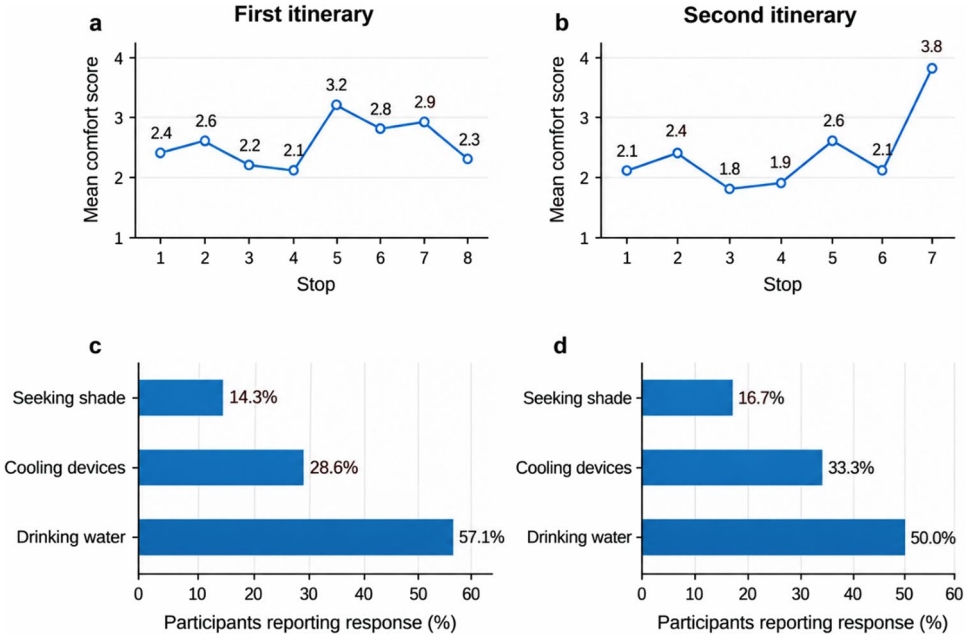


Fig. 5 Comparison of comfort scores and personal adaptation measures for both itineraries: (**a**) mean comfort scores by stop for the first itinerary; (**b**) mean comfort scores by stop for the second itinerary; (**c**) personal adaptation measures reported during the first itinerary; and (**d**) personal adaptation measures reported during the second itinerary. Source: Authors' questionnaire results

Qualitative results

The data collected from the two questionnaires also yielded counterintuitive results, further questioning the variety of heat perceptions and heat-adaptation reflexes among the participants, as well as how they problematized heat within the social and spatial dimensions of the transitioning district.

First result: Heat perception is never pure, fully transparent, and easily accessible to an individual. It is filtered by other senses and by a broader valuation of place. The first questionnaire provided insights into a dimension that participants were not fully aware of at the beginning of the walk, namely, the synesthetic approach that connects thermal senses with visual and acoustic information. It is well known (Zhou et al., 2022; Pellerin and Candas 2003; Li et al. 2025) that the landscape, in its holistic dimensions, affects thermal comfort.

In our first questionnaire, 14 participants responded on a 4-point Likert scale regarding noise perception, with none selecting the 'pleasant' category, indicating broadly negative noise experiences (52% unpleasant, 37% bothering, and 11% acceptable). Noise levels measured ranged from approximately 56 dBA at Gambetta Plaza (Stop 5, first itinerary) to highs of 73–74 dBA during the first itinerary around the Rousseau-Lénine avenues intersection (Stop 1, itinerary 1) and the Vaillant-Couturier bus stop (Stop 3, itinerary 1), reflecting high acoustic noise in these areas. Sound impacts heat perception: Boulevard Vaillant-Couturier is described as dirty and oppressively hot, although no visible litter was present: "It's full of cars, with old neglected buildings, trash strewn about, and pavement and asphalt everywhere you look… It feels very hot in summer—and throughout the year too." (Participant 1, Walk 1). Participants' accounts reflected how noise and heat are strongly intertwined in people's sensory experiences. "The Vaillant Couturier bus stop area becomes very hot due to the asphalt, and the cars exacerbate

the issue. The noise adds to the discomfort, especially during peak hours. Sometimes it calms down a bit, and a breeze passes through since the space is large, but the degraded buildings make it an eyesore." (Participant 7, Walk 1). And "The traffic in Lenin Street is constant. There are many shops and restaurants, and I don't enjoy being there because of the continuous noise from people and cars. Except for a few trees, it feels noisy and hot, even more so because of the noise, even though we're close to the Seine." (Participant 12, Walk 1).

Visual characteristics of public space affect spatial meanings and overall well-being. First, our research demonstrates that people tend to prioritize visual cues over other environmental factors. This suggests that when a place appears visually normal and acceptable, with moderate activity levels and minimal signs of degradation, occupants generally do not report high thermal discomfort scores or feel the need for specific heat adaptation strategies. The Gambetta area illustrates this phenomenon well. This open space is located within the renovation zone and is surrounded by high-rise buildings. It is intended to accommodate a future metro station. Its openness to informal leisure activities is characteristic of the transitory use of public space. Gambetta is also part of urban developers' efforts to "greenify" Ivry. The site is officially designated as the "Gambetta Lawn" and is intended to accommodate sports activities, but residents refer to it as "Gambetta Beach" (Fig. 6a). Barbecues and picnics are common, and social gatherings are frequent. The site has thus become a symbol of conviviality among Ivry-Port's diverse populations. The ground consists of a barren sandy expanse with sparse spontaneous vegetation. Constant traffic noise comes from the busy roundabout adjacent to the site. Consequently, Gambetta ranks among Ivry's hottest and dirtiest locations. Yet, against all expectations, the Gambetta area shows high positive alliesthesia scores (see Fig. 6b). Despite these nuisances, Gambetta elicits highly positive feelings and functions as a vibrant "third space," as its representation as a leisure beach evokes an imaginary of pleasant heat associated with holidays, leisure, and relaxation. Heat is felt but is not perceived as a problem by residents.

Second result: Thermal walks attentive to the complexities of lived space can reveal conflicts over access to urban resources, among which access to cool spaces is becoming increasingly important. This became particularly apparent at one of the flagship public spaces of the Ivry-Port redevelopment project. The thermal walk led participants to Minoterie Square (Stop 6, Itinerary 1), a former street that the municipality transformed into a pedestrian public space. The square is surrounded by recently constructed four-story residential buildings occupied by newcomers. The square offers a secluded and comfortable public space. It features a line of open water, numerous trees, manicured gardens, benches, a children's playground, and even a free drinking water fountain. No doubt, participants confirmed that the spot is among the coolest; it has a great visual appeal and offers a comfortable experience, whether for those who visit it for their own purposes or for those who spend a relaxing break outdoors. For these reasons, Minoterie Square is an attractive and always bustling area, especially during the day, when it is full of people, including children and youngsters who play in the playground or play football, as well as retired individuals who enjoy listening to the radio and music (Fig. 7a). Residents socialize in the shade and on benches not only during the day. When the weather is nice, especially when it is hot, they also enjoy the site on weekends, late in the evening, and at night. Participants are aware that Minoterie is the site of a tense conflict between

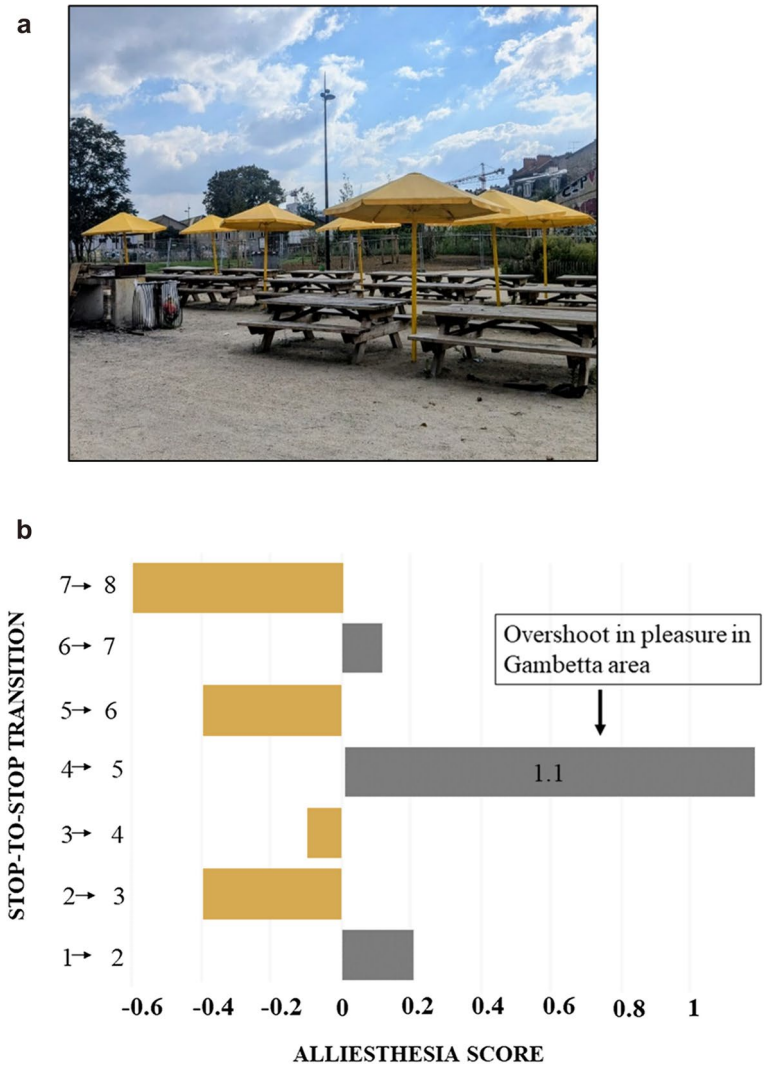


Fig. 6 **a** Gambetta area photography. Source: Authors. **b** Alliesthesia score of itinerary 1. Source: Authors' analysis of questionnaire results

residents of nearby buildings and users of the complete street (Fig. 7b). The residents hang panels on the building façades, denouncing the disturbance that the users of public space create below their windows and balconies. Acknowledging this context, participants expressed a poor valuation of the place. This illustrates that climate planning itself does not necessarily produce positive heat sensations or a feeling of comfort and ease. This also reminds us of the interference between personal and collective memories and heat perception. Individuals' responses and interpretations were shaped by their prior knowledge of city life (Nikolopoulou et al. 2001). For example, one participant described Square Minoterie as noisy and uninviting, even though no children were present during the walk: "I never really considered sitting there. I didn't even notice how fresh it was because the noise and children playing around captured my attention… Whenever someone mentions Square Minoterie, I immediately think of the noise. (Participant 2, Walk 2).

The striking contrasts between the Gambetta and Minoterie spots have sparked reflections that reach well beyond the immediate scope of collaborative thermal diagnosis.

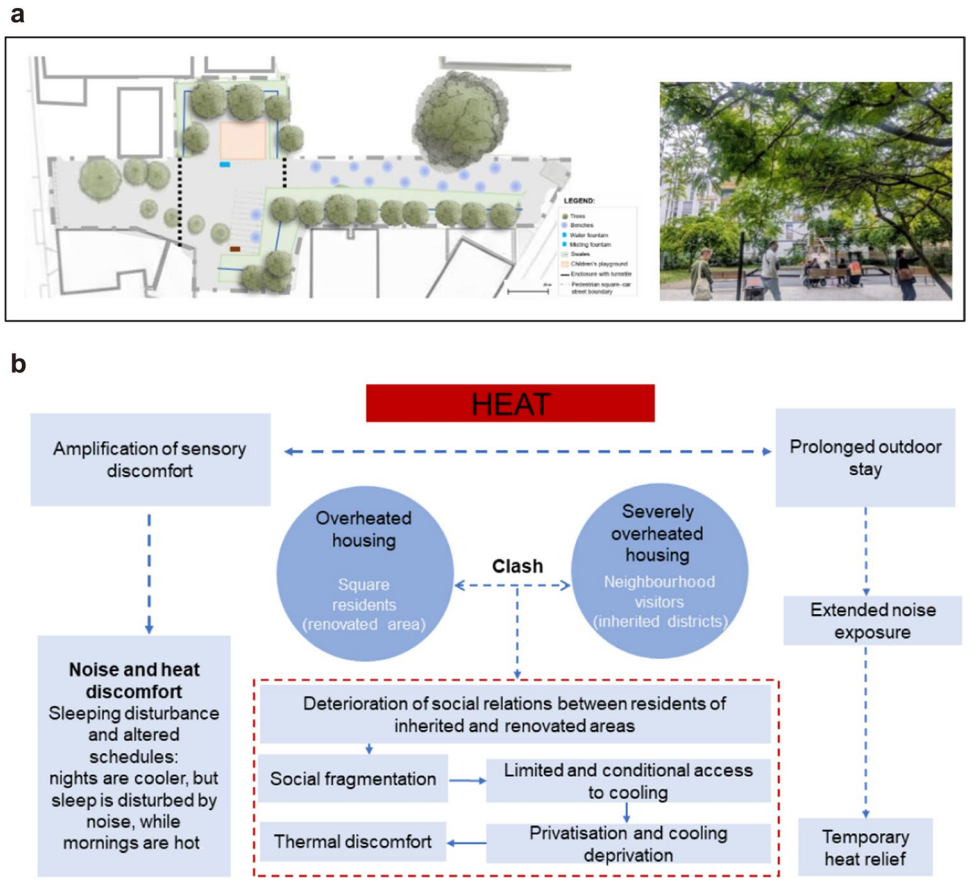


Fig. 7 **a** Minoterie square layout and photography. Source: Authors. **b** How heat becomes a social conflict. Source: AuthorsSource: Authors' questionnaire results.

The findings may appear counterintuitive to climatologists, yet they reveal social and spatial dynamics at the core of urban planning's complexities. Perceived heat does not always correspond to measured climatic conditions. Minoterie, though objectively cooler, is often experienced as crowded, noisy, or even oppressive, and its favorable thermal characteristics, shaped by both urban form and microclimate, tend to go unnoticed. In contrast, the Gambetta Area is warmer in both climatic and social terms, fostering a sense of vitality that makes thermal discomfort less pronounced. These insights prompt broader reflection on the aims, substance, and implications of urban experiments that address heat adaptation and thermal comfort in disadvantaged urban contexts.

Part 3. Urbanizing heat: the learning potential of thermal walks

This part links the core knowledge established in the first part with the account of the experiments presented in the second. It offers an ex-post review of the case study and further elaborates on the conditions for a successful thermal walk and the learning outcomes it generates. Yet, success relies on the aims defined by thermal walk organizers and the implicit meanings attached to experimentation. By introducing a ternary typology of experimentalist practices (Ansell and Bartenberger 2016), we reveal both limitations and significant opportunities for integrating thermal walks into planning processes.

Targeting and engaging diverse participants in thermal walks

Climate walks benefit from a diverse group of participants, although diversity itself requires careful consideration. Climate-oriented thermal walks are among the types of experiments known as controlled experiments. Controlled experimentation in the natural sciences seeks to draw valid inferences about cause and effect by isolating pertinent factors, treating them as variables in the phenomena being explored. Controlled experiments aim to minimize bias (either originating from the researcher or from the survey participants). When certain cause-and-effect mechanisms are identified and other factors are controlled, the results are considered valid and reproducible in other experiments.

Climatologists consider that working with a heterogeneous group enriches the experiment but makes survey conduct quite difficult, whereas homogeneity helps manage mobile groups and makes the results more reliable in terms of factor controllability. Heterogeneity refers to recruiting people from diverse backgrounds, with varied opinions and thought styles. Homogeneity, when applied to urban climatology, refers to individuals with sufficient communication skills, a readiness to learn, and the capacity to invest time and energy (Van de Kerkhof and Wieczorek 2005). In this approach, the social dimensions related to class, race, living conditions, and access to urban resources tend to become blurred. This is not a problem in itself, as climatologists are merely seeking a variety of physiological perceptions and body-mind thermal associations. Yet it is a crucial question for urban-planning-oriented thermal walks. Homogeneity is a trap if it refers to easy-to-recruit and easy-to-manage participants, the kind of participatory public that responds to recruitment campaigns. In our case, it points toward the most affluent population, which has recently occupied the renovated part of the district. Such individuals have developed specific ways of coping with heat. They enjoy new homes with adequate ventilation, balconies, and communal gardens within the blocks. Such individuals are well-informed about climate risks. And, effectively, we had no difficulty enrolling the community collective Urbanicc, which brings together such climate-conscious citizens. The Thermal Walk project met the collective's intention to raise awareness about environmental problems and negotiate solutions with authorities regarding climate and greening issues. Yet we were very aware of the need to include the district's more disadvantaged residents.

Large and vulnerable families can be found in degraded, overcrowded flats. This population struggles for survival and may not respond rapidly to such an experiment, but suffers from a lack of fresh air and access to open, public green spaces. Hence, the participation of this other part was essential not only to address varied heat perceptions but also for democratic reasons. In this disadvantaged district, with support from a citizen-science-oriented municipal office, participants' perceptions were not simply tested. The walk served an educational aim, fostering climate awareness and empowerment. The group, unfamiliar with thermal questions and lacking training to articulate their perception of heat, initially struggled to engage. Although they possessed vernacular heat knowledge, they were unaccustomed to the terminology used in the test. Their early responses showed a conformist tendency. Gradually, they were guided to move beyond perplexity, uncertainty, and helplessness in response to certain questionnaire items. They came to see sensory experience as trainable and improved their perceptual awareness by freely handling thermal cameras and other tools. Through the walk,

vulnerable participants gained access to climate knowledge that middle-class groups already possessed.

Learning (more) by doing (more) thermal walks, an opportunity to root climate experiments

To examine this learning-by-doing approach, we investigated heat perceptions and adaptation through two thermal walks conducted in the contrasting physical and social settings of Ivry-Port. We also engaged with local community organisations and the municipality to explore how this experimental approach could be adapted to local conditions.

Future thermal walks are planned across different seasons, days of the week, and times of day, as the initial two walks took place on Saturday and Wednesday afternoons for participants' convenience. We also plan walks tailored for children and youth to simulate school and leisure mobility in the district, as well as walks for active adults commuting to transit hubs. Additionally, we envision thermal events substituting walking with biking, leveraging the Greater Paris public bike network. This would allow longer itineraries, including visits to the district's southern edge, where a newly inaugurated park remains inaccessible by other means.

Such an iterative approach operates at two levels. We can progressively refine the walk design as we develop a better understanding of what can be learned from each thermal walk. In this perspective, the thermal walk may be considered as an experimental niche. The public-, spatiotemporal-, and mobility-driven variations of the event will produce a range of best (and worst) attempts. Such a modest yet repetitive approach is key to localizing the thermal walk within a specific urban context, where the physical and social idiosyncrasies of Ivry-Port's urban heat will be delineated.

Politicizing thermal walks: realizing their disruptive potential

Thermal walks may encourage people to engage more critically with heat in their everyday environments and serve as catalysts for social, behavioral, or political change. It depends on the type of protocol and how it balances a descriptive perspective with a solution-oriented approach. It also varies depending on how the thermal walk is conducted. In a controlled setting, the process is largely constrained by the predefined protocol. In contrast, a generative approach to participatory thermal experimentation makes unexpected observations and interactions part of the process, fostering deliberation and creativity.

When describing the Ivry-Port thermal walk, we discovered two counterintuitive results. One of the warmest places was "not so warm" and not so negatively perceived (Gambetta was even labeled by deprived inhabitants as their "beach"). The place marked by the greatest social conflict was also the coolest, yet few participants felt at ease there. In such cases, the value of the thermal walk went beyond documenting microclimatic conditions and assessing participants' perceptions. The discrepancies between the two became a starting point for collective discussion. Together with the community's pressure on local officials, the thermal walk fostered a broader process of deliberation that developed in several directions, including bringing the issue onto the municipality's agenda.

The Minoterie conflict, which emerged during the thermal walk as both a heat-related and social issue, was part of a broader local controversy. Noise disturbance, crowding, and conflicts between different uses of the square had also been raised by residents and local collectives (Dubois 2024). These counterintuitive findings show how the thermal walk can move beyond documenting thermal conditions to generate new questions and possible explanations. Rather than confirming an expected relationship between measured heat and perceived discomfort, participants proposed explanations for the discrepancies they encountered. At Minoterie, the unexpected discomfort reported in one of the coolest places was associated by participants with noise, crowding, and conflicts between different uses. In this sense, the reasoning was abductive: an unexpected observation led to the formulation of plausible explanations rather than simply confirming an expected correspondence between measured heat and perceived discomfort (Hanson 1958; Douven 2025).

Such experiments may also provide a means of justification for urban authorities. It is possible that officials allowed this project to proceed because they sought to verify and legitimize the relevance of public-space redevelopment efforts. They may have hoped that the thermal walk would serve, through a form of participatory scientific experimentation, as proof of the quality of their planning. Yet, with sites such as Fauconnières block, Gambetta Plaza, and Minoterie square, among others, this thermal walk opened up a series of "black boxes".

When the thermal walk protocol is properly designed to facilitate dialogue, it can naturally trigger a virtuous process that generates new ways of addressing the complex issue of perceived heat. This awareness-raising dimension is also found in Barbosa and Klok (2020), who present the thermal walk as an experiential learning tool. As previously shown in Table 1, their approach has a potentially politicising dimension because it connects individual thermal experiences with collective reflection on the built environment.

Participants not only ask themselves questions about urban heat but also pose them to one another. Discussions arise not only about hot and cold, but also about why certain places are perceived as problematic or not. A thermal walk conceived within a generative experimental perspective is thus as much a political act as a scientific one. In the socially and racially divided urban context of Ivry-Port, it can create opportunities for dialogue across social groups and contribute to climate preparedness. In this sense, the task is not merely to uncover the relationship between thermal experience and the major sociodemographic transition currently underway in this Parisian suburb, which is undergoing managed gentrification to avoid displacing vulnerable populations. The challenge of the generative thermal walk is to help residents, collectives, and local officials move forward together in imagining new forms of civic practice. It thus helps create "climate citizens," in the sense proposed by MacGregor (2006), who act beyond individual interest and take responsibility for their collective future.

Proposals for grounding thermal walk protocols in urban transformation

Table 3 synthesizes some of the shortcomings identified in existing thermal walk practices and the ways in which they can be addressed through methodological refinement. Based on the literature reviewed, these challenges are organized across four operational stages (design, preparation, implementation, and post-walk follow-up). The proposed improvements are consistent with an open and adaptive approach that requires

Table 3 Limitations and recommendations for thermal walk improvement

Part of the experiment	Flaws of current thermal walk practices	Recommendations for improvement
1. Design phase Conception of thermal walks: *involving scholars and public stakeholders.* Launching the project with the community and public actors. Agreeing on the types of thermal walks and how they can influence public policies. Defining the target area and timeline.	Lack of integration with urban planning processes. Lack of flexibility in survey methods	Enhance collaboration by building stronger partnerships between researchers and local urban public actors to promote integration and cooperation. - Prefer online questionnaires distributed to participants at each stop, enabling asynchronous participation without requiring everyone to be present simultaneously. - Allow flexibility in the walk's structure to encourage spontaneous participant observations that reveal valuable local knowledge.
2. Preparatory phase Recruitment of participants and preliminary information shared: Introducing and explaining the concept of a thermal walk. Demonstrating the use of scientific instruments. Discussing the itinerary.	Lack of cultural and social context in thermal perception analysis. Limited participant involvement in the process and data collection	Investigate how social categories, gender and culture influence thermal perception as well as local practices and sensitivities. Participants are not just there to answer questions; they need to gain real added value from their involvement. This means recognizing their contributions, providing restitution sessions to help them understand the findings, and involving them in data collection by using scientific instruments, so that they can be active collaborators rather than data collectors.
3. Implementation	Managing experiment fatigue Focusing on walking dynamics without sufficient emphasis on stationary states	A short questionnaire is preferable. While a Likert scale can be interesting, it tends to standardize responses into fixed categories, which limits the range of perceptions and may lead participants to answer without much reflection. Extending investigations to include thermal perceptions during both stationary and active phases, in order to understand how urban design supports these states differently
4. Follow-up to the Thermal Walk, **post–Thermal Walk phase** - Data analysis and interpretation. - Discussions and actions initiated by the Thermal Walk.	Overemphasis on quantitative data, climate modeling, and weather data. Disconnection between objective and subjective information.	Use a multisensory and phenomenological approach to urban walks and heat. Developing hybrid data visualization methods like thermal notations. Visualizing qualitative data in sensory mapping.

continuous adjustment and moves beyond controlled protocols toward iterative and generative experimentation.

Conclusion

The article develops an analytical and critical approach to participatory thermal experiments by combining a literature review with a reflexive analysis of an experimental protocol conducted in a deprived suburban district of Paris. The first part clarifies the epistemological foundations of the thermal walk concept, tracing its origins through a review of the literature and its subsequent incorporation into the social sciences. Thermal walks call for approaches that go beyond individual perspectives, recognizing heat not only as an individual experience but also as a social reality that affects communities at multiple levels. The transfer of thermal walks from climatology to urban planning reflects a broadening of their objectives, from the observation and assessment of thermal conditions to the inclusion of participation, collective learning, and action research.

Thermal walks can now be situated within a broader family of mobile and participatory methods. The second part presents an experiment aimed at highlighting the importance of residents' engagement in understanding and contributing to urban heat planning. It produced counterintuitive results that were analyzed through both climatic and critical urban and social perspectives and discussed during and after the walks to foster debate and shared solutions. The third part examined the extent to which thermal walks can enhance both collective and individual awareness of the dynamic relationships among bodies, minds, and urban spaces. The main finding is that thermal walks, when conceived as iterative and generative experiments, foster collective learning. Experimentation can, in this sense, operate as a potentially disruptive process that questions not only the sensory and psychological dimensions of heat but also the social order and existing climate-planning practices.

The Ivry-Port experiment also demonstrates the critical capacity of the thermal walk. Walking through and collectively interpreting these spaces revealed a gap between official representations of public space and residents' situated experiences. Urban redevelopment increasingly associates public spaces with narratives of greening, even where these environmental qualities remain limited or ambiguous. Such representations accompany broader processes of urban revalorization. Residents, however, do not necessarily evaluate these spaces primarily through their environmental or thermal qualities. The case of Gambetta is particularly revealing in this respect: a space that is objectively among the hottest and least green can nevertheless generate positive thermal perceptions because of its social uses, conviviality, and association with leisure. Thermal walks, therefore, make it possible to critically examine not only whether an urban space is hot or cool, but also who qualifies it as such, according to which criteria, and with what implications for urban transformation.

Finally, the Ivry-Port experiment shows how thermal walks can evolve from one-off experiments into tools for public policy. Iterative thermal walks can help build practical expertise, generate local knowledge, and support institutional learning. In Ivry-Port, the experience gained through these experiments can inform the intermunicipal climate plan, particularly its efforts to address heat vulnerability and adaptation in deprived neighborhoods. The value of thermal walks, therefore, extends beyond the knowledge produced during a single event. Over time, they can help bring residents' experiences into dialogue with experimental practices and local climate policy.

Under these conditions, thermal walks connect the production of knowledge with collective action. They contribute to the development of climate citizens while providing a means of questioning established categories of thermal comfort, environmental quality, and climate adaptation. Their contribution to urban experimentation lies precisely in this dual capacity: to reveal discrepancies between planned and lived thermal environments and to turn these situated observations into resources for collective and policy action.

Author contributions
Work repartition:-Z.H. intervened in the following tasks: Figures, tables.- B.N. intervened in the following tasks: Supervision.- Both authors were involved in the following tasks: writing and editing.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with relevant national and institutional guidelines. Ethics committee approval was not required under the applicable regulations and institutional policies. Informed consent was obtained from all participants by the research team prior to their participation in the study.

Competing interests

H. Zouad is simply an author. N.Baron is a co-author and a guest editor of this issue.

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